
Where Does Platonic Convergence Live? Pointwise Curvature of a Galaxy Embedding Manifold and What a Linear Map Can See

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Abstract

1 The Platonic Universe tested the Platonic Representation Hypothesis (PRH) by
2 measuring the convergence of foundation-model representations of galaxy images
3 with mutual k -nearest-neighbour agreement (MKNN). We ask whether that conver-
4 gence is concentrated where the embedding manifold is flat. A pointwise curvature
5 field from a trained decoder, validated against analytic truth, finds mean-curvature
6 magnitude correlating negatively with cross-survey neighbour agreement at every
7 latent dimension, and that local density removes most of it at $d = 20$, half at $d = 25$
8 and little of the already small $d = 32$ value. A second-order expansion of a linear
9 map restricted to the manifold explains why: a readout sees the second fundamen-
10 tal form contracted with its normal component, $\langle w_N, \Pi \rangle$, and, at second order, an
11 alignment’s error splits into a connection mismatch and the relative second funda-
12 mental form projected into the target normal space; mean curvature retains only
13 the metric trace of Π and determines neither. On a unit-normalized embedding that
14 Hessian splits into a sphere term, proportional to the readout’s own value, and a
15 shape term. On a synthetic surface the sphere term holds at -0.74 to -0.80 across
16 samplings that swing the $\|H^S\|$ partial from -0.29 to $+0.26$; on 86,471 labelled
17 galaxies the mismatch between the label Hessian and the full probe-facing Hes-
18 sian is negative wherever the label has local structure and survives cross-fitting,
19 the sphere term is negative on two of four labels under a weak-ridge sensitivity,
20 and the shape term alone takes a label-dependent sign consistent with that decom-
21 position. Mean curvature of a single representation is therefore not a viable expla-
22 nation for weak Platonic convergence; the obstruction is relative and, for the fitted
23 global linear alignment, already present at first order.

24 1 Introduction

25 The Platonic Representation Hypothesis [Huh et al., 2024] holds that models trained on different data
26 converge toward a shared representation; Duraphe et al. [Duraphe et al., 2025] tested it on the sky:
27 galaxies imaged by two surveys (Legacy Survey and Hyper Suprime-Cam, HSC) and embedded by
28 a family of different vision models share only 0.3 to 2.3 percent of their nearest neighbours across
29 the two surveys. We ask where on the manifold that agreement lives: if two models disagree most
30 where the embedding surface bends sharply, local mean curvature should associate negatively with
31 local cross-survey agreement.

32 We contribute (i) a pointwise mean-curvature and second-fundamental-form instrument from a
33 trained decoder, validated against analytic truth in the production regime; (ii) a density-controlled
34 test of the curvature–alignment prediction, which finds no effect free of density; (iii) the residual
35 expansion of a linear map on a submanifold, which names the curvature such a map responds to and

shows mean-curvature magnitude is blind to it; and (iv) known-surface and real-data tests on the readout side, where the expansion’s terms behave as predicted.

2 Pointwise curvature from a decoder

Data. DINOv3 [Siméoni et al., 2025] ViT-B/16 embeddings of 101,725 galaxies imaged by both HSC [Aihara et al., 2018] and the Legacy Survey [Dey et al., 2019], (Platonic Universe release [Dupaphe et al., 2025]); we draw 10,000 rows once with a recorded seed. A second set of 86,471 single-survey ViT-B embeddings with catalogue labels (r -band magnitude, photometric redshift, smooth fraction, stellar mass; the AstroPT galaxy set [Smith et al., 2024]) serves the readout experiment. Embeddings are unit-normalized (data on S^{767}). Curvature is evaluated only where J has full rank.

Instrument. Let $F : \mathbb{R}^d \rightarrow \mathbb{R}^{768}$ be the decoder of a plain auto-encoder (three hidden layers of width 250, SiLU), fit on 80% of rows; curvature is evaluated on held-out rows. With $J = DF(z)$, metric $g = J^\top J$, normal projector $P_N = I - Jg^{-1}J^\top$ and second fundamental form $\Pi = P_N D^2 F(z)$, the mean-curvature vector of the raw decoder surface is $H_{\text{raw}} = \text{tr}_g \Pi$, by automatic differentiation. On the unit sphere every d -submanifold carries a radial component of H equal to $-d$ that says nothing about shape. To isolate shape, the map differentiated is $F/\|F\|$, whose H splits exactly as $-d\hat{x} + H^S$ ($H_{\text{rad}} = -d$ to 10^{-14}); H^S , normal to M within the sphere, carries the shape. Section 3 uses the pre-registered $\|H_{\text{raw}}\|$; Sections 4–5 use the projected decoder. Decoder curvature has served as a regularizer [Lee and Park, 2023] and been read from templates [Acosta et al., 2022]; here it is a validated instrument.

Validation. Scored against closed-form curvature on rank (Spearman), direction (cosine) and magnitude ratio. At $D = 768$ an in-sphere surface $G(z) = Q \text{ normalize}([\phi(z); 0.8h(z); 0 \dots])$ (ϕ inverse stereographic $\mathbb{R}^{16} \rightarrow S^{16}$, h a scalar sum of four Gaussian bumps, Q a rotation; 18-dimensional span) at $n = 86,471$, $k = 2,048$ gives rank 0.94 against the true $\|H^S\|$ (equivalently $\|H\|$, the radial part being fixed), direction cosine 0.999 and magnitude ratio 1.00 for the full vector (rank 0.75 under 25% noise).

3 Curvature and cross-survey convergence

Probe. Per-point mutual k -nearest-neighbour score $\text{MKNN}_i = |N_k(z_1)_i \cap N_k(z_2)_i|/k$ at $k = 20$ between the HSC and Legacy Survey embeddings, the Platonic Universe statistic made local, correlated with $\|H_{\text{raw}}\|$ over all 10,000 points at $d \in \{20, 25, 32\}$. Centered kernel alignment [Kornblith et al., 2019] is the second probe.

Result. The raw association $\rho(\|H_{\text{raw}}\|, \text{MKNN})$ is -0.112 , -0.128 and -0.024 at $d = 20, 25, 32$ ($p < 10^{-5}$, $< 10^{-5}$, 0.009), in the predicted direction. But k -NN statistics are density-sensitive [Radovanović et al., 2010]: local density (the $k = 30$ nearest-neighbour estimate $k/(nV_d r_k^d)$ on the Legacy embedding, $d = 20$) correlates with agreement at -0.21 at every d and with $\|H_{\text{raw}}\|$ at $+0.43$, $+0.32$ and $+0.01$, so residualizing on density leaves -0.024 , -0.066 and -0.022 : 78%, 49% and 8% of the association removed. Under a permutation null within density strata only $d = 25$ survives ($p < 5 \times 10^{-5}$; three decoder seeds; the CKA tertile gap within strata agrees at $d = 20$ and 25). Removing the radial component of H_{raw} (the raw surface is not on the sphere, so the two rank at $\rho = 0.89$ – 0.96 rather than identically) collapses the $d = 25$ partial to -0.023 , into the range of the two d that fail. Convergence is not detectably concentrated by mean-curvature magnitude once density is controlled.

4 What a linear map sees on a curved manifold

Let $M \subset \mathbb{R}^D$ be a d -dimensional submanifold, $d < D$; at x_0 take an orthonormal tangent frame J , the second fundamental form Π (normal-valued, symmetric $d \times d$), $H = \text{tr} \Pi$, and for a vector w its tangent and normal parts $w_T + w_N$. In normal coordinates u at x_0 , M is $x_0 + Ju + \frac{1}{2}\Pi(u, u) + O(|u|^3)$, so

$$[\nabla_M(w \cdot x)]_J = J^\top w, \quad \text{Hess}_M(w \cdot x) = \langle w_N, \Pi \rangle, \quad \Delta_M(w \cdot x) = \langle w, H \rangle. \quad (1)$$

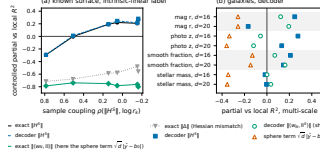


Figure 1: (a) Known surface, intrinsically linear label, exact geometry: three-control partial against local R^2 as the sample's density-curvature coupling is set by construction (γ from -1 to 1). (b) Galaxies, 512 anchors, $k = 2,048$: partial under the multi-scale density control for the decoder's $\|H^S\|$ (filled squares), the shape term (open circles) and the sphere term (open triangles).

To second order, a linear function is as non-linear on M as M bends in that function's normal direction [Lee, 2018]. For $M \subset S^{D-1}$ the normal bundle splits into the sphere's radial direction and the rest, $\Pi = \Pi^S - g \hat{x}$, so $\text{Hess}_M(w \cdot x) = \langle w_N, \Pi^S \rangle - (w \cdot \hat{x}) g$ (2): a linear readout of a unit-normalized embedding is bent by the manifold's shape and, whatever the manifold, by the sphere in proportion to the readout's own value $\hat{y} - b_0$. For a label y on M , a global affine probe $\hat{y} = w \cdot x + b_0$ and a patch $u \sim N(0, s^2 I_d)$ in normal coordinates, embedding and label locally C^4 (the $\frac{1}{2}$ is the Gaussian constant; real k -NN patches share only the tensor structure), with $\bar{c}$ the patch-mean residual, $\delta = [\nabla_M y]_J - J^\top w$, $\Delta = \text{Hess}_M y - \langle w_N, \Pi \rangle$,

$$\mathbb{E}_{\text{patch}}[(y - \hat{y})^2] = \bar{c}^2 + s^2 \|\delta\|^2 + \frac{1}{2} s^4 \|\Delta\|_F^2 + O(s^4 \|\delta\|) + O(s^6). \quad (3)$$

For two embeddings F, G of the same objects in a common chart u and a linear map A with $AJ_F = J_G$, the same expansion gives $G - (G_0 + A(F - F_0)) = \frac{1}{2} [J_G(\Gamma^G - \Gamma^F) + \Pi_G - A \Pi_F](u, u) + O(|u|^3)$, with Γ the induced connections: the second-order obstruction to linear alignment is a connection mismatch plus the *relative* second fundamental form, and reduces to the latter when the induced connections agree at x_0 (in particular under a local intrinsic isometry); the extrinsic term's normal part is $\Pi_G - P_N^G A \Pi_F$.

So the curvature a linear readout pays for is $\langle w_N, \Pi \rangle$, and an alignment's second-order error has an extrinsic part whose normal component is $\Pi_G - P_N^G A \Pi_F$; $\|H\|$ contains only the norm of the metric trace of Π and need not correlate with either; the second-order term scales as s^4 , and with fixed- k neighbourhoods the sampling density sets s , so curvature effects scored on patches are density-sensitive; and when the label has intrinsic curvature of its own, Δ carries the residual and the curvature piece can take either sign.

5 The probe-facing curvature, on a known surface and on galaxies

Statistic: five-fold out-of-fold ridge probe [Alain and Bengio, 2016] at the pre-registered $\alpha = 100$ from embedding to label; local R^2 inside each of 512 anchors' 2,048-neighbourhoods; rank-partial Spearman against local R^2 with Freedman-Lane p -values [Freedman and Lane, 1983], controlling for log patch radius, local label variance and count and, stronger, log radius at $k \in \{16, \dots, 2048\}$ (five scales). Per anchor we compute $\|H^S\|$ and the probe-facing Hessian $\|\langle w_N, \Pi \rangle\|_g$ with $w_N = w - Jg^{-1}J^\top w$, split by Eq. (2) into its shape and sphere terms.

Known surface. The validation surface is resampled with weight $\|H^S\|^\gamma$, so only where the points sit changes (Figure 1a). The $\|H^S\|$ partial follows the sampling from -0.29 to $+0.22$, and the decoder reproduces it. The probe-facing Hessian $\|\langle w_N, \Pi \rangle\|$ has partial correlation -0.74 to -0.80 at every sampling. On this surface it is the sphere term of Eq. (2) (rank 0.996–1.000 with $\sqrt{d}|\hat{y} - b_0|$; in-sphere part 20 to 40 times smaller). With a nonlinear label the sphere term is indistinguishable from zero and $\|\Delta\|$ carries the partial (-0.2 to -0.6), as Eq. (3) predicts.

Galaxies. Here the two terms of Eq. (2) trade places: the shape term dominates the probe-facing Hessian (rank 0.96–0.99 with the full norm). We estimate $\text{Hess}_M y$ per anchor by a least-squares quadratic of the label on the tangent-projected coordinates $u = g^{-1}J^\top(x - x_0)$ of its 2,048 neighbours (tangent-plane projection is a local Monge chart, in which the first metric derivatives and hence the Christoffel symbols vanish at the anchor, so the quadratic coefficient is the covariant Hessian there). Three regularities (Table 1; other decoder seeds and widths in Appendix A).

label	$\ H^S\ $		shape $\ \langle w_N, \Pi^S \rangle\ $		sphere $\sqrt{d} \hat{y} - b_0 $		mismatch $\ \Delta\ $		alignment	
	$d=16$	$d=20$	16	20	16	20	16	20	16	20
mag_r	+0.25	-0.17	+0.12	+0.19	-0.26	-0.26	-0.39	-0.40	+0.35	+0.43
photo_z	+0.28	+0.13	-0.06*	-0.12	-0.33	-0.36	-0.45	-0.44	+0.26	+0.24
smooth_fraction	+0.21	+0.15	+0.07*	+0.16	-0.13	-0.15	-0.09*	-0.17	+0.19	+0.07*
stellar_mass	-0.02*	-0.00*	+0.00*	+0.03*	-0.35	-0.35	-0.04*	-0.09*	+0.01*	+0.02*

Table 1: Galaxies, 512 anchors, $k = 2,048$: rank-partial Spearman against local R^2 under the multi-scale density control; alignment is $\cos_g(\text{Hess}_M y, \langle w_N, \Pi \rangle)$. * not significant at 0.05; the three-control partials agree in sign in every significant cell.

The sphere term is negative on every label and both d (-0.13 to -0.36) under the sealed probe; with a weak-ridge probe ($\alpha = 1$, Appendix B) it stays negative for magnitude and smooth fraction (-0.21 , -0.12) and vanishes for redshift and stellar mass, where the sealed-probe sign came with ridge shrinkage of extreme predictions. Wherever the label has local structure (magnitude, redshift, smooth fraction; stellar mass is indistinguishable from zero throughout) the mismatch is negative (-0.09 to -0.45 ; unchanged under cross-fitting, Appendix B) and the alignment $\cos_g(\text{Hess}_M y, \langle w_N, \Pi \rangle)$ positive ($+0.07$ to $+0.43$). Writing $K = \langle w_N, \Pi \rangle$, curvature lowers the second-order residual relative to $K = 0$ iff $2\langle \text{Hess}_M y, K \rangle > \|K\|^2$; positive alignment is the first-order condition (bending toward the label’s curvature helps until it overshoots); the mismatch column is the full test. The shape term alone is label-signed (positive for magnitude and smooth fraction, null to weakly negative for redshift), consistent with Eq. (3): it is one of three terms in $\|\Delta\|^2 = \|\text{Hess}_M y\|^2 - 2\langle \text{Hess}_M y, \langle w_N, \Pi \rangle \rangle + \|\langle w_N, \Pi \rangle\|^2$, so raising it shrinks or enlarges the mismatch according to that condition; empirically its sign follows the cross term’s where that is nonzero and the remaining square where it is null. $\|H^S\|$ changes sign between $d = 16$ and 20 for magnitude, as on the curve of Figure 1a. The same regression on the readout $w \cdot x$ returns the point cloud’s own $\langle w_N, \Pi \rangle$ (median cosine with the decoder’s 0.55 – 0.65), and an independent split-half quadratic-chart estimator of Π gives the same shape-term sign on three of four labels.

Across encoders. The quadratic-chart estimator’s mean-bending trace on five encoders at the same anchors (ViT-B, DINOv3, CLIP-B, ConvNeXt-B, ViT-L; $d = 16$, $k = 2,048$) has density-controlled associations with the magnitude probe’s local R^2 of -0.24 , 0.00 , $+0.27$, $+0.23$ and -0.19 : a trace statistic has no encoder-stable sign, consistent with Eq. (1).

6 Discussion

A decoder yields a pointwise second fundamental form at production scale that agrees with an independent estimator and with the data. The residual expansion names what a linear map responds to, and on galaxies its terms behave as predicted: the Hessian mismatch carries the predicted negative sign, robust to cross-fitting and probe strength; the sphere term does so on two of four labels under a weak-ridge sensitivity; and the shape term is label-signed because it enters through its alignment with, and magnitude relative to, the label’s own Hessian. The same expansion explains why mean-curvature statistics disagree across estimators and encoders: they are traces of a tensor that each map contracts differently. If weak convergence is attributed to the failure of a linear alignment map, one embedding’s mean curvature is not the relevant second-order quantity; after tangent alignment the relevant normal term is $\Pi_G - P_N^G A \Pi_F$, which needs both embeddings’ second fundamental forms and a map between tangent frames. Our pilot at $d = 20$ (2,048 anchors, two seeds; Appendix C) finds the fitted global ridge map carries HSC tangent frames into the Legacy embedding with 44% residual, a first-order mismatch, and no estimator-stable second-order association: the global-decoder, locally aligned and decoder-free estimates read $+0.04$ (n.s.), -0.18 to -0.23 and $+0.13$, disagreeing in sign. In sum: single-representation mean curvature is not the second-order statistic linear alignment depends on; a readout responds to $\langle w_N, \Pi \rangle$, and alignment additionally to tangent mismatch, connection mismatch and relative extrinsic bending. On this survey pair, single-model mean curvature has no robust density-controlled association with cross-survey convergence.

Limitations. One survey pair. The label’s Hessian is a local estimate (R^2 gain over linear 0.10 – 0.17); its norm is reliably ranked (split-half ρ 0.92) but its direction only moderately (tensor cosine 0.19 –

0.35). Stars are per cell; mismatch and alignment are the primary hypotheses, the rest exploratory. Permutation p -values assume anchor exchangeability under overlapping patches.

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Appendix A: decoder ablations

Table 2 repeats the four quantities of Table 1 at $d = 16$ under the multi-scale density control for two further decoder initialisation seeds and for a decoder of width 400^3 instead of 250^3 (all fits reach variance explained 0.952). Every significant sign of the main table is retained.

Appendix B: sensitivity of the mismatch and sphere columns

Cross-fitting: each 2,048-patch is split at random into halves; $\text{Hess}_M y$ is fitted on one half and local R^2 scored on the other (both directions shown), so the Hessian estimate and the outcome share no rows. The split-half cosine between the two Hessian estimates is the reliability of the estimate. Weak ridge: the probe refit at $\alpha = 1$ instead of 100 (global out-of-sample R^2 rises by 0.10 to 0.15), which substantially reduces the shrinkage of extreme predictions; the sphere term is recomputed from that probe.

label	quantity	seed 0	seed 1	seed 2	400 ³
mag_r	shape	+0.12	+0.05*	+0.12	+0.05*
	sphere	−0.26	−0.26	−0.27	−0.26
	mismatch	−0.39	−0.35	−0.34	−0.34
	alignment	+0.35	+0.31	+0.37	+0.41
photo_z	shape	−0.06*	−0.09*	−0.09	−0.06*
	sphere	−0.33	−0.35	−0.34	−0.35
	mismatch	−0.45	−0.43	−0.42	−0.41
	alignment	+0.26	+0.24	+0.25	+0.25
smooth_fraction	shape	+0.07*	+0.20	+0.15	+0.10
	sphere	−0.13	−0.15	−0.14	−0.11
	mismatch	−0.09*	−0.05*	−0.08*	−0.07*
	alignment	+0.19	+0.14	+0.09	+0.10
stellar_mass	shape	+0.00*	−0.06*	+0.05*	+0.09
	sphere	−0.35	−0.35	−0.35	−0.35
	mismatch	−0.04*	+0.03*	+0.02*	+0.03*
	alignment	+0.01*	−0.02*	+0.03*	−0.02*

Table 2: Decoder ablations, $d = 16$, 512 anchors, multi-scale density control. * not significant at 0.05.

label	d	mismatch, cross-fit		alignment, cross-fit		Hess. split cos	sphere, $\alpha = 1$
		fit A/score B	fit B/score A	fit A/score B	fit B/score A	p50	(main: $\alpha = 100$)
mag_r	16	−0.38	−0.34	+0.33	+0.31	+0.35	−0.21 (−0.26)
mag_r	20	−0.43	−0.40	+0.39	+0.37	+0.24	−0.22 (−0.26)
photo_z	16	−0.42	−0.36	+0.23	+0.20	+0.32	+0.05* (−0.33)
photo_z	20	−0.42	−0.42	+0.22	+0.22	+0.20	+0.03* (−0.36)
smooth_fraction	16	−0.10	−0.17	+0.17	+0.19	+0.33	−0.12 (−0.13)
smooth_fraction	20	−0.15	−0.21	+0.05*	+0.03*	+0.23	−0.13 (−0.15)
stellar_mass	16	+0.01*	−0.04*	−0.03*	+0.06*	+0.31	−0.02* (−0.35)
stellar_mass	20	−0.06*	−0.07*	+0.01*	−0.00*	+0.19	−0.02* (−0.35)

Table 3: Cross-fitted mismatch and alignment partials (multi-scale control), split-half reliability of the label Hessian, and the sphere-term partial under a weak-ridge probe.

Appendix C: relative second fundamental form, pilot

Two sphere-projected decoders (HSC = F , Legacy = G ; Phase 7 protocol), a global ridge map A from x_F to x_G (fit on the 8,000 training rows), 2,048 seeded anchors. Columns: holdout R^2 of A ; median first-order obstruction $\|AJ_F - J_G L\|/\|AJ_F\|$ with $L = J_G^+ AJ_F$; then multi-scale density-controlled partials against MKNN ($k = 20$; log radius at $k \in \{10, 30, 100, 300\}$ in both spaces) of each decoder’s $\|H^S\|$, the first-order obstruction, $\|\Pi_G(L, L) - P_N^G A \Pi_F\|$, the same with L fitted on 256 neighbours’ latent codes, and a decoder-free estimate from the quadratic coefficient of the alignment residual on F ’s tangent coordinates.

run	R_A^2	tan. resid	$\ H^S\ $ HSC	$\ H^S\ $ Legacy	tan. resid	Π_{rel}	Π_{rel} (local L)	Π_{rel} (data)
$d = 20$, seed 0	0.64	0.44	−0.15	+0.04*	−0.18	+0.04*	−0.18	+0.13
$d = 25$, seed 0	0.64	0.43	−0.10	+0.06	−0.16	+0.05	−0.23	+0.13
$d = 20$, seed 1	0.64	0.44	−0.11	+0.05	−0.19	+0.04	−0.20	+0.14

Table 4: Relative-II pilot. * not significant at 0.05.